Iterative Frequency Domain Channel Estimation for DFT-Precoded OFDM Systems using In-Band Pilots

Chan-Tong Lam *Student Member*, David D. Falconer *Life Fellow*, and Florence Danilo-Lemoine *Member*

***Abstract*—We consider two techniques of in-band frequency domain multiplexed (FDM) pilots using interleaved frequency domain multiple access (IFDMA) signal with a Chu sequence for DFT-precoded OFDM (or single-carrier (SC)) systems. One, called frequency domain superimposed pilot technique (FDSPT), superimposes pilot tones onto scaled or deleted data tones, which preserves spectral efficiency at the expense of a slight performance loss. The other, called frequency expanding technique (FET), multiplexes pilot tones by displacing data tones, which slightly reduces spectral efficiency. Using FDM pilots in SC systems facilitates flexible and efficient assignment of signals to available spectrum. We propose an iterative frequency domain decision-directed interference cancellation technique to reduce the intersymbol interference level of SC signals with FDSPT pilots (resulting from the suppression of data tones). Moreover, we propose a low complexity frequency domain iterative decision-directed channel estimation (IDDCE) technique for SC systems using FDM pilots. Using IDDCE, the frame error rate (FER) performance for coded SC systems using FET and FDSPT pilots with interference cancellation is found to be about 0.2 dB and about 0.5 dB, respectively, away from the FER performance with known channel frequency response at FER=10^{-2}. FDSPT pilots can also be used for OFDM systems with channel coding. It is found that an extra 1 dB of SNR is required at FER=10^{-2}, compared with that using the conventional FET pilots for OFDM systems.**

***Index Terms*—In-band pilots, single-carrier, OFDM, DFT-precoded OFDM, iterative frequency domain equalization, frequency domain channel estimation.**

I. Introduction

NEXT generation wireless systems will likely use flexible combinations of frequency domain block transmission methods such as orthogonal frequency domain multiple access (OFDMA) and single carrier with frequency domain equalization (SC-FDE). For example SC may be preferred for the uplink of cellular systems because of its low peak to average power ratio (PAPR) and the resulting power amplifier efficiency in the user terminal. Using the generalized multicarrier approach [11], a SC signal can be generated by Discrete Fourier Transform (DFT)-precoding an OFDM signal. Hence, an SC system can also be called a DFT-precoded OFDM system. Channel estimation using pilot (training) symbols is employed in current 3rd generation (3G) [1] and will likely be employed in proposed future beyond-3G wireless communication systems [2], [3]. Although bandwidth-consuming, channel estimation using pilot symbols remains attractive in practice because of the decoupling of symbol detection from channel estimation, which reduces complexity and relaxes the required identifiability conditions [4], [5]. The pilot symbols for single-carrier (SC) systems are traditionally time multiplexed within or in between discrete Fourier transform (DFT) blocks [6], [7]. When time multiplexed (TDM) pilot symbols are inserted periodically to track time-varying channels, it is known as pilot symbols assisted modulation (PSAM) [4], [5], [8]. The pilots for orthogonal frequency division multiplexed (OFDM) systems are usually frequency domain multiplexed (FDM) within the data bandwidth, resulting in a more efficient and flexible assignment of signals to available spectrum. In the OFDM context it is sometimes called pilot-assisted channel estimation (PACE) [9], [10]. With the generalized multicarrier approach (GMC) [11], a SC signal can also be generated by DFT-precoding an OFDM signal, i.e. the DFT of the data symbols are used instead of the data symbols before the inverse DFT (IDFT) operation [12]. Hence, SC systems generated in this way can also be called DFT-precoded OFDM systems. Generating SC signals in the frequency domain avoids conventional complex symbol-by-symbol filtering and affords an extra degree of spectrum flexibility.

In this paper we extend the use of FDM pilots to SC systems using the GMC signal generation approach. Aiming for application in broadband mobile channels, we frequency-multiplex multiple in-band pilot tones, along with data tones, using the concept of an interleaved frequency domain multiple access (IFDMA)[1] signal [14] with a Chu sequence [15], which has constant envelope and uniform spectrum.

Two techniques of FDM pilots for SC systems are considered here. One is to superimpose pilot tones onto scaled or deleted data tones, and is termed frequency domain superimposed pilot technique (FDSPT). Early work on the insertion of pilot symbols in the frequency domain for SC systems can be found in [16], [17], called transparent tone-inband (TTIB), in which a single reference tone is inserted in the middle of the signal bandwidth using a notch filter along with suitable mixing and filtering in the passband and it is intended for flat fading mobile communication systems. A "data dependent superimposed training" (DDST) technique is introduced in [18], [19], and in general form, in [20], to distort the data symbol sequence (in the time domain) to create periodic frequency nulls for insertion of pilot tones. The main difference of our FDSPT technique with parameter $\alpha = 0$

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C.-T. Lam is currently with the Bureau of Telecommunications (DSRT) of Macao S.A.R., China (e-mail: chantonglam@gmail.com).

D. Falconer is with Dept. of Systems and Computer Eng., Carleton University, Ottawa, ON, Canada (e-mail: ddf@sce.carleton.ca).

F. Danilo-Lemoine is with Dept. of Systems and Computer Eng., Carleton University, Ottawa, ON, Canada (e-mail: fdanilo@sce.carleton.ca).

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[1]IFDMA has had many names, such as FDOSS [13].

(defined later) from those of [18], [19] and [20] is that it is defined and implemented in the frequency domain instead of the time domain. It is also applicable to pilots with arbitrary spacing in the frequency domain. Furthermore, in contrast to the above references, FDSPT and FET (described in the next paragraph) are combined here with iterative block equalization, with cancellation to suppress intersymbol interference caused by FDSPT, with iterative decision-directed channel estimation and with forward error correction coding. Other previous works on superimposed pilots include [21], [22] and [23], in which periodic pseudo-random pilots are added to data, and are extracted for channel estimation at the receiver by time-averaging over multiple periods.

The other main pilot technique considered here is to shift groups of data frequencies for multiplexing of the pilot tones, called frequency expanding technique (FET), a technique usually employed by OFDM systems, where it is called PACE (pilot-assisted channel estimation) [9], [10]. In this paper, the generation of the transmitted SC signals with FET pilots is in the frequency domain. Moreover, the general training and precoding framework of [20] includes both FET and FDSPT, but is based on a specific channel estimator (DFT channel estimator). For FET pilots, there is no limitation on the specific channel estimator used. We also consider the peak-to-average power ratio (PAPR) effects resulting from the FET pilots. FET pilots do not result in intersymbol interference but create pilot overhead, which lowers the spectral efficiency. [2]

With initial channel estimates using pilot symbols, tentative data decisions (re-modulated data symbols) can be used as *extended pilots* to improve the channel estimates, thus improving the overall system performance [24]–[26]. The tentative decisions can be made either from the output of the equalizer or from the channel decoder. Using decisions from the equalizer requires smaller processing delay and lower implementation complexity at the expense of error propagation at relatively higher SNR, than using decisions from the channel decoder. When an *a posteriori* (APP) channel decoder is available, joint maximum a posteriori (MAP) detection and channel estimation (optimal in terms of minimizing BER) is possible [27]. However, implementation complexity of the optimum joint MAP data detection and channel estimation is still prohibitive, as the trellis expansion induced by the channel estimation task in the decoder enhances greatly the complexity of the conventional forward/backward procedure [27]. Hence, we are interested in suboptimal receivers with separated processes of data detection and channel estimation for low complexity. In [26], a time domain decision-directed channel estimator is incorporated into the iterative process of a single-carrier frequency domain equalizer to improve the channel estimate, using TDM pilots. The channel estimation involves the inverse of a matrix with size of the decisions of a block, which is complex to implement. In [24], the authors propose a low complexity iterative filtering technique using the decisions from the channel decoder to improve the channel estimates for SC systems using TDM pilots in time-varying flat fading channels. This technique is modified in [25] for OFDM systems in time-varying frequency selective fading channels. With the aid of pilot symbols, the initial channel estimates are obtained, in the frequency domain, using a cascaded 2×1 dimensional (2×1D) FIR Wiener filter [10]. Using the initial channel estimates, the log-likelihood ratio (LLR) of the coded symbols can be calculated using an APP decoder. These LLRs are then used to obtain soft decisions for the channel estimation in the next iteration. In this paper, we modify the frequency domain channel estimator using a filtering technique for OFDM systems (described in [25]) so that it can be used for SC systems with FDM pilots. We propose a low complexity frequency domain iterative decision-directed channel estimation (IDDCE) technique for SC systems in time-varying frequency selective fading channels. Instead of using a more complex APP decoder, a Viterbi decoder is used. Moreover, we consider simple hard decisions from the decoder and the equalizer.

Our contributions in this paper include the analysis of FDM pilots for *SC systems* using the GMC approach combined with a frequency domain decision-directed interference cancellation technique for SC signals with FDSPT pilots. Moreover, we propose a low complexity IDDCE technique for SC systems using FDM pilots. We also obtain the results for OFDM systems with FDM pilots for comparison. We emphasize that the novelty of the paper is to use the in-band FDM pilots and iterative frequency domain decision-directed techniques, which are typically used for OFDM systems, for SC or DFT-precoded OFDM systems. The FET pilot technique is an integral part of the uplink wide area DFT-precoded OFDM transmission mode in the EU WINNER Project on new radio interfaces [2]. The evolving 3GPP LTE (long term evolution) standards group also proposes DFT-precoded OFDM (called "localized SC-FDMA") for the uplink, but with pilot signals that are time-multiplexed with data [3].

The rest of the paper is organized as follows: Sec. II provides the description of system models, including background on the generation of SC signals with FDM pilots. Sec. III presents and analyzes SC signals with FDM pilots, which also includes the proposed iterative decision directed interference cancellation technique for SC signals with FDSPT pilots. Sec. IV presents the proposed frequency domain IDDCE for SC systems using FDM pilots in detail. Sec. V provides the simulation results and discussions, followed by the conclusions in Sec. VI.

II. System Model

Fig. 1 shows the block diagram of the transmitter and receiver using GMC signal generation. A transmitted block, containing data and pilots, consists of L samples plus a L_{cp} cyclic prefix, which is assumed to be longer than the known channel impulse response length.

The complex data symbol a_m has zero mean and unit variance. We assume a size-M data plus pilot symbol block transmission ($M < L$). The data tones A_ℓ (M-point FFT of $\{a_m\}_{m=0}^{M-1}$ for FDSPT and $(M-N)$-point FFT of $\{a_m\}_{m=0}^{M-N-1}$ for FET) and pilot tones P_ℓ (N-point FFT of a N-length Chu sequence [15]) are multiplexed into a single frequency domain sequence, denoted as X_ℓ , padded with

[2] The spectral efficiency here is simply defined as the ratio of the number of data symbols to the total number of data plus pilot symbols per block (or per OFDM symbol).

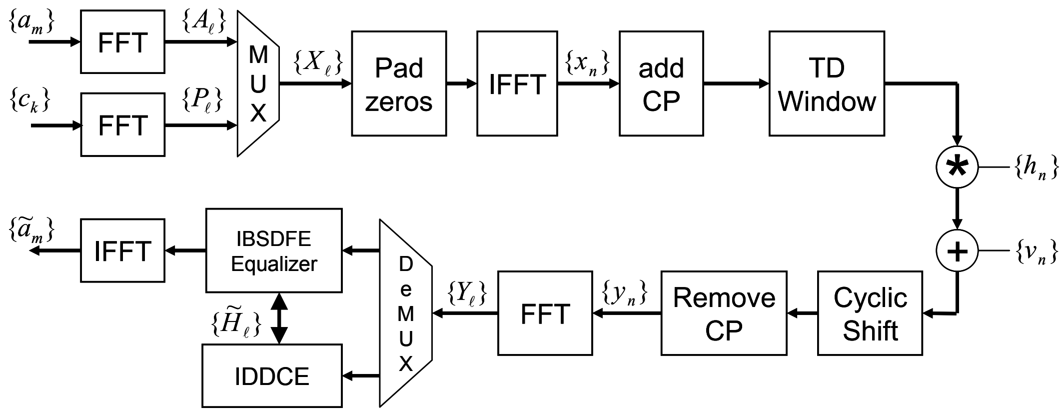

Fig. 1. GMC Transmitter and Receiver Structure with Channel Estimation using In-band Pilots

zeros to length L, as described in detail in the next section. Note that $N < M < L$ and for OFDM system the data tones A_ℓ are simply the data symbols, i.e. $A_\ell = a_\ell$, for $\ell = 0, 1, \ldots, M-1$ for FDSPT pilots and $\ell = 0, 1, \ldots, M-N-1$ for FET pilots. The N pilot tones are interspersed among data tones at equal intervals (displacing data in the case of FET and replacing data or superimposed on data in the case of FDSPT). The transmitted waveform is the inverse FFT (IFFT) of the composite sequence X_ℓ. The waveform containing only the equidistant pilot tones $\{P_\ell\}$ can be regarded as an IFDMA signal [13], [14]. By padding enough zeros to $\{X_\ell\}$ for a total length of L, the transmitted signal with an oversampling factor of L/M can be obtained by taking the L-point IFFT. After adding the cyclic prefix (CP) to prevent inter-block interference, the signal can be time-domain-windowed to reduce the side-lobes of the transmitted spectrum.

In the receiver side, before removing CP, a cyclic shift of the received samples is necessary due to the rolloff of the time window skirt. Taking the L - point FFT of the received baseband sample y_n and then removing the last $L-M$ frequencies, we obtain the received pilot and data tones

$$Y_\ell = X_\ell H_\ell + V_\ell, \qquad \ell = 0, 1, ..., M-1 \tag{1}$$

where H_ℓ and V_ℓ represent the channel response at frequency ℓ and the ℓth frequency noise samples, respectively. For FDSPT, X_ℓ is the ℓth element of

$$\mathbf{X}_S = [\beta P_0 + \alpha A_0, A_1, ..., \beta P_1 + \alpha A_K, A_{K+1}, ..., A_{M-1}] \tag{2}$$

where α and β are the scaling factors for the data and pilot tones at pilot locations, respectively, P_ℓ is the ℓth pilot tone and K is the pilot spacing. In this paper we assume $\alpha^2 + \beta^2 = 1$.[3] For FET, X_ℓ is the ℓth element of

$$\mathbf{X}_E = [P_0, A_0, ..., P_1, A_{K-1}, ..., P_{N-1}, ..., A_{M-N-1}] \tag{3}$$

Note that N more data symbols can be sent using FDSPT. The equalizer used is the frequency domain iterative block soft decision feedback equalizer (IBSDFE) [28]. Its iterative process does not include decoding or channel estimation. The frequency domain output of the IBSDFE equalizer at the ith iteration is given as [28]

$$U_\ell^{(i)} = W_\ell^{(i)*} Y_\ell + F_\ell^{(i)*} \bar{A}_\ell^{(i-1)}, \qquad \ell = 0, 1, \ldots, M-1 \tag{4}$$

where * denotes complex conjugate, $\bar{A}_\ell^{(i-1)}$ is the FFT of soft feedback symbols at the $(i-1)$th iteration, $W_\ell^{(i)}$ is the ℓth feedforward filter tap, given as [4]

$$W_\ell^{(i)} = \frac{H_\ell \left(S_\ell^{(i)} - \mu^{(i)} \right)}{|H_\ell|^2 S_\ell^{(i)} + \sigma^2}, \qquad \ell = 0, 1, \ldots, M-1 \tag{5}$$

where $S_\ell^{(i)}$ is the conditional mean squared difference between data symbols and their soft decision counterparts at the ith iteration, σ^2 is the noise variance and $\mu^{(i)}$ is the Lagrange multiplier, given as

$$\mu^{(i)} = -\frac{\sum_{\ell=0}^{M-1} \frac{\sigma^2}{|H_\ell|^2 + \sigma^2}}{\sum_{\ell=0}^{M-1} \frac{|H_\ell|^2}{|H_\ell|^2 + \sigma^2}}. \tag{6}$$

The ℓth feedback filter tap at the ith iteration is given as

$$F_\ell^{(i)} = 1 - H_\ell^* W_\ell^{(i)}, \qquad \ell = 0, 1, \ldots, M-1. \tag{7}$$

One or two iterations are generally sufficient to yield a significant performance improvement over linear equalization on severely dispersive channels. It is more effective than a conventional hard decision DFE, and avoids its error propagation problem, at the expense of slightly higher complexity. Note that when the feedback filter is not used, the IBSDFE is equivalent to a linear minimum mean square error (MMSE) equalizer.

Given the received frequency samples at the pilot frequencies $Y_{\ell'}$ ($\ell' \in \mathbf{\Phi} \triangleq \{0, K, \cdots, (N-1)K\}$), the frequency domain IDDCE produces the estimated channel frequency response $\tilde{H}_\ell$, which is interpolated to all frequencies and then input to the IBSDFE. The channel estimation is performed over a frame, which contains Q consecutive blocks. Finally, the estimated data symbols $\tilde{a}_m$ are obtained by taking either M or $(M-N)$-point IFFT of the estimated data tones.

[3]Note that for the proposed iterative channel estimation technique using FDSPT pilots in this paper, we only consider the case of $\alpha = 0$ for simplicity of the removal of pilots and equalization process as we want to propose a low complexity iterative frequency domain channel estimation technique. In cases of $\alpha \neq 0$, the FFT of the estimated data symbols appears to be the interference for channel estimation, while imperfect channel estimates and data decisions result in performance degradation of equalization (see [29]). A more complex algorithm can be used to enhance the performance at the expense of higher implementation complexity.

[4]Note that in this paper we normalize the equalizer taps, i.e. $\frac{1}{M} \sum_{\ell=0}^{M-1} W_\ell^{(i)*} H_\ell = 1$. Our formulation of IBSDFE is equivalent to the soft detection version of [28], but the order in which normalizations are performed differs slightly.

III. In-band FDM Pilots for DFT-precoded OFDM Systems

A. Frequency Domain Superimposed Pilot Technique (FD-SPT)

The idea of FDSPT is to periodically scale frequencies for superimposing of the IFDMA pilot tones. The advantage is not to expand the signal bandwidth, thus maintaining the spectral efficiency. However, as will be shown later, it suffers performance degradation due to the loss of part of the useful data frequencies and induces slightly higher peak to average power ratio (PAPR) than when no pilots are present. [5] We want to obtain an expression for the signal samples before adding CP samples, denoted as x_n in Fig. 1 and analyze the implication of periodically frequency scaling and superimposing in terms of PAPR. Define a frequency-scaling window as

$$Q_\ell = \begin{cases} \alpha, & \ell \in \mathbf{\Phi} \\ 1, & \ell \notin \mathbf{\Phi} \end{cases} \tag{8}$$

where $0 \leq \alpha \leq 1$. The baseband transmitted signal (with IFDMA training signal) x_n can then be expressed as

$$x_S(n) = \underbrace{\frac{1}{L}\sum_{\ell=0}^{M-1} A_\ell Q_\ell e^{j\frac{2\pi n\ell}{L}}}_{distorted\ data\ signal} + \underbrace{\frac{\beta}{L}\sum_{k=0}^{N-1} P_k e^{j\frac{2\pi nkK}{L}}}_{IFDMA\ training\ signal} \tag{9}$$

where $0 \leq \beta \leq 1$ and the subscript $_S$ represents the signal for FDSPT. The $\{P_k\}$ are the DFT coefficients of a Chu sequence. It is shown in [13] that the resulting waveform is equivalent to a single carrier waveform. The properties of the Chu sequences are such that the frequency domain coefficients $\{P_k\}$ and the corresponding time domain sequence have constant magnitude. The second term in (9) represents an IFDMA single carrier waveform added to a FDSPT, FET or OFDM signal. Therefore, for the purpose of comparing the PAPR performance among the signals using these techniques, we focus on the first term in (9). It can be shown that the envelope fluctuation is largest when $\alpha = 0$. The transmitted signal with $\alpha = 0$ and without the pilot tones can be found as

$$x'_S(n) = \frac{M'}{L}\sum_{m=0}^{M-1} a_m g_S\left(n - \frac{mL}{M}\right) \tag{10}$$

where $g_S(x) = \frac{\mathrm{sinc}\left(\frac{x}{I}\right)}{\mathrm{sinc}\left(\frac{x}{L}\right)} \frac{\mathrm{sinc}\left(\frac{K-1}{L}x\right)}{\mathrm{sinc}\left(\frac{K}{L}x\right)} e^{\frac{j\pi x}{I}}$ is the sampled impulse response of a channel with periodic nulls, $M' = M - N$ and $n = 0, 1, ..., L-1$. The high sidelobes of the $g_S\left(n - \frac{mL}{M}\right)$ pulse increase the PAPR.

FDSPT pilots can also be used for OFDM systems with channel coding. Since a data symbol is represented as a subcarrier in OFDM system, the removal of data tones for pilots is equivalent to *puncturing* in the context of channel coding.[6] Hence, unlike SC system, the performance loss for OFDM systems with FDSPT pilots results only from the *puncturing* operation. Moreover, FDSPT pilots do not induce significantly higher PAPR than OFDM signals without pilots, due to the already high PAPR of OFDM.

[5] In this paper, we focus on the low complexity iterative frequency domain channel estimation technique for DFT-precoded OFDM systems using in-band FDM pilots. More detailed discussions on the issue of PAPR for DFT-precoded OFDM signals can be found in [29].

[6] In most of this paper, we assume the scaling factor $\alpha = 0$.

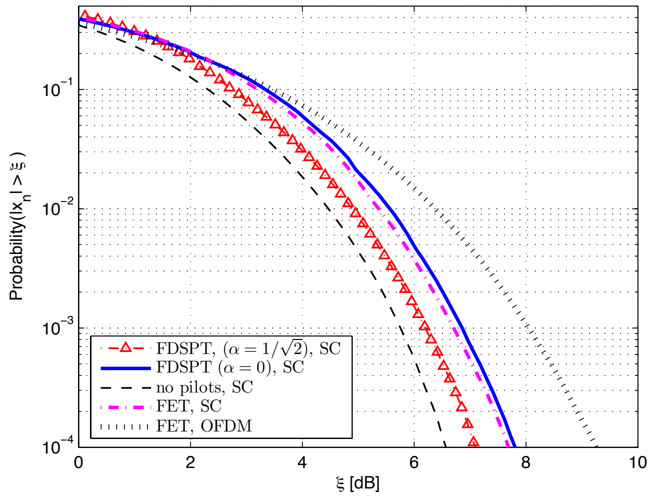

Fig. 2. Comparison of CCDF of Magnitude of SC Signals with FDSPT and FET Pilots ($M = 1024, N = 256, I = 5, \eta = N/M = 25\%$)

B. Frequency Expanding Technique (FET)

FET evenly expands groups of frequencies for multiplexing of IFDMA pilot tones. FET has slightly lower spectral efficiency than that of the conventional SC signal without pilots due to the expansion of data frequencies to accommodate the pilot tones. Note that FET is the frequency domain pilot technique commonly used in OFDM systems, where it is often called pilot assisted channel estimation (PACE) [10]. When using FET, the multiplexed data and pilot tones are as defined in (3), where pilot tones P_ℓ are periodically multiplexed with the data tones A_ℓ. The baseband transmitted samples x_n in Fig. 1 is given as

$$x_E(n) = \underbrace{\frac{1}{L}\sum_{\substack{\ell=0 \\ \ell\notin\mathbf{\Phi}}}^{M-1} X_\ell e^{j\frac{2\pi n\ell}{L}}}_{distorted\ data\ signal} + \underbrace{\frac{1}{L}\sum_{k=0}^{N-1} P_k e^{j\frac{2\pi nkK}{L}}}_{IFDMA\ training\ signal} \tag{11}$$

where the subscript $_E$ represents the signal for FET. Similar to the FDSPT case, we consider the first term in (11), which can be further expanded as

$$x'_E(n) = \frac{M'}{L} e^{j\frac{\pi n}{I}} \sum_{m=0}^{M'-1} a_m g_E(m,n) e^{-j\pi\left(\frac{(M'-1)m}{M'}\right)} \tag{12}$$

where $g_E(m,n) = \frac{\mathrm{sinc}\left(N\left(Kn/L - m(K-1)/M'\right)\right)}{\mathrm{sinc}\left(Kn/L - m(K-1)/M'\right)} \times \frac{\mathrm{sinc}\left((K-1)\left(n/L - m/M'\right)\right)}{\mathrm{sinc}\left(n/L - m/M'\right)}$, $M' = M - N$ and $n = 0, 1, ..., L-1$. It is obvious that the time-varying modified pulse shaping filter $g_E(m,n)$ results in higher PAPR than a SC system without multiplexing pilot tones. $g_E(m,n)$ is no longer a pure *sinc* filter, allowing more data symbols to be summed to produce possible high PAPR. Fig. 2 shows the complementary cumulative distribution function (CCDF) of the magnitude of the transmitted signal x_n, where x_n is as defined in Fig. 1. At CCDF$= 10^{-4}$, the FDSPT with $\alpha = 0$ and FET have about 1.5 dB advantage over OFDM and 1.5 dB disadvantage over SC modulated signal without pilot tones. The higher the value of α is, the better the PAPR.

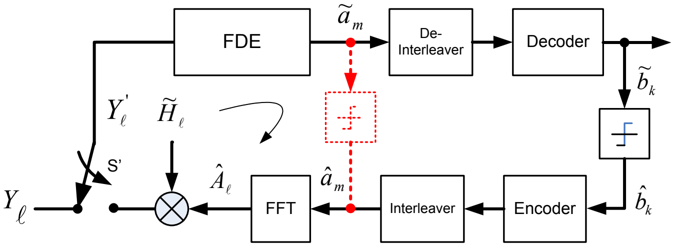

Fig. 3. Block Diagram of the Proposed Decision-directed Interference Cancellation Technique for SC Signals with FDSPT Pilots (FDE: frequency domain equalizer)

The main difference between FET pilots and FDSPT pilots is the lower spectral efficiency but no performance degradation. We will show that the performance degradation resulting from the removal of data tones for pilot tones can be compensated by using channel coding and decision-direction interference cancellation.

C. Decision-directed Interference Cancellation for SC Signals with FDSPT Pilots

The removal of the data tones for superimposing of the pilot tones results in BER performance degradation. The first term in (9) can be expressed as

$$x'_S(n) = a'_n - \frac{(1-\alpha)\eta}{I}\sum_{m=0}^{M-1} a_m \frac{\text{sinc}\left(\frac{n}{I}-m\right)}{\text{sinc}\left(K\left(\frac{n}{L}-\frac{m}{M}\right)\right)} \cdot e^{j\pi\left(\frac{n}{L}-\frac{m}{M}\right)K(N-1)} \quad (13)$$

where $\{a'_n\}_{n=0}^{L-1}$ is the oversampled data samples without frequency scaling, $\eta = N/M$ denotes the pilot overhead ratio and $I = L/M$ is the oversampling factor. The second term of (13) can be considered as the intersymbol interference term due to the scaled data tones, which results in BER performance degradation. It is obvious that the interference level decreases with α and increases with η.

Given the estimated channel frequency response, denoted as $\tilde{H}_\ell$, and the DFT of the data decisions $\{\hat{a}_m\}$[7] either from the frequency domain equalizer or the decoder output, denoted as $\hat{A}_\ell$, we can *compensate* for the removal (or scaling) of the data tones by *re-constructing* the received frequency samples at the pilot frequencies $\mathbf{\Phi}$, as shown in Fig. 3. After the initial iteration, Switch S' is at the left hand side for $\ell \notin \mathbf{\Phi}$ and at the right hand side for $\ell \in \mathbf{\Phi}$. The *compensated* received frequency samples, denoted as Y'_ℓ, can be obtained as

$$Y'_\ell = \begin{cases} \tilde{H}_\ell \hat{A}_\ell & \ell \in \mathbf{\Phi} \\ H_\ell A_\ell + V_\ell & \ell \notin \mathbf{\Phi} \end{cases} \quad (14)$$

where V_ℓ is the ℓth noise frequency sample. Notice that the difference between Y'_ℓ and Y_ℓ (see (1)) is that for $\ell \in \mathbf{\Phi}$, Y_ℓ contains the *received* frequency samples for the transmitted pilot tones, while Y'_ℓ contains the received frequency samples for the transmitted data tones. Moreover, to obtain Y'_ℓ, the DFT of the data decisions and channel frequency response are needed. Note that the ℓth received frequency sample for a block without pilots, denoted as $\bar{Y}_\ell$, is given as

$$\bar{Y}_\ell = H_\ell A_\ell + V_\ell, \qquad \ell = 0, 1, \ldots, M-1 \quad (15)$$

where M is the number of data symbols per block.

The objective of the interference cancellation algorithm for SC signal with FDSPT pilots is to obtain Y'_ℓ, using $\hat{A}_\ell$ and $\tilde{H}_\ell$, such that $Y'_\ell \approx \bar{Y}_\ell$ for $\ell = 0, 1, \ldots, M-1$. Let $\tilde{H}_\ell = H_\ell + \Delta H_\ell$ and $\hat{A}_\ell = A_\ell + \Delta A_\ell$, where ΔH_ℓ and ΔA_ℓ are defined as the channel estimation and the decision errors at the ℓth frequency, respectively. Y'_ℓ can be re-written as

$$Y'_\ell = \begin{cases} H_\ell A_\ell + V'_\ell & \ell \in \mathbf{\Phi} \\ H_\ell A_\ell + V_\ell & \ell \notin \mathbf{\Phi} \end{cases} \quad (16)$$

where V'_ℓ is the interference due to the channel estimation and data decision errors, given as

$$V'_\ell = \Delta H_\ell \Delta A_\ell + \Delta H_\ell A_\ell + H_\ell \Delta A_\ell. \quad (17)$$

Comparing (16) with (15), we notice that the difference between Y'_ℓ and $\bar{Y}_\ell$ is the noise term at the pilot frequencies $\mathbf{\Phi}$. Note that in the absence of channel estimation and data decision errors, the received frequency samples at $\mathbf{\Phi}$ are noiseless. In this case, the performance of the frequency domain equalizer would be at least as good as that without the FDSPT data frequency nulls.

The performance improvement is less significant when IBSDFE (instead of linear MMSE) equalizer is used, resulting from the interference cancellation capability of the feedback filter within the IBSDFE equalizer. Consider the equalization of the received frequency samples at $\mathbf{\Phi}$ for SC signals with FDSPT signals ($\alpha = 0$) without the proposed interference canceller. Assuming channel frequency nulls at $\mathbf{\Phi}$, i.e. $H_\ell = 0$ for $\ell \in \mathbf{\Phi}$ and using (4), it can be shown that

$$U_\ell^{(i)} = \bar{A}_\ell^{(i-1)}, \qquad \ell \in \mathbf{\Phi}, \quad (18)$$

where $\bar{A}_\ell^{(i-1)}$ is the ℓth frequency of the DFT of the feedback soft decisions from the $(i-1)$th iteration. When a linear equalizer is used, we have

$$U_\ell^{(i)} = 0, \qquad \ell \in \mathbf{\Phi}. \quad (19)$$

We see that the IBSDFE equalizer provides essentially the same degree of interference cancellation capability for SC signals with FDSPT pilots. With the proposed interference canceller, we have

$$U_\ell^{(i)} = W_\ell^{(i)*} Y'_\ell + F_\ell^{(i)*} \bar{A}_\ell^{(i-1)}, \qquad \ell \in \mathbf{\Phi}, \quad (20)$$

where $W_\ell^{(i)}$ and $F_\ell^{(i)}$ are defined in (5) and (7) respectively. Comparing (18) and (20), we see the difference that feedforward and feedback filter are applied on the received frequency samples Y'_ℓ and $\bar{A}_\ell^{(i-1)}$ when the proposed canceller is used. Note that this difference is applicable only at the pilot frequencies $\mathbf{\Phi}$. The IBSDFE equalizer with decision directed interference cancellation technique at $\mathbf{\Phi}$ is almost equivalent to a version of IBSDFE in which the feedforward filter is after the feedback and subtraction operation, i.e. the feedforward filter is within the feedback loop of the equalizer. The only differences are (1) in IBSDFE with interference cancellation, hard decisions are subtracted, instead of soft decisions; and (2)

[7]Notations: $\widetilde{[\ ]}$ is used to denote estimated channel frequency response after the W 2×1 D Wiener filter or the soft value of data symbols. $\widehat{[\ ]}$ is used to denote the raw estimates for the W 2×1D Wiener filter or the hard decision of data symbols.

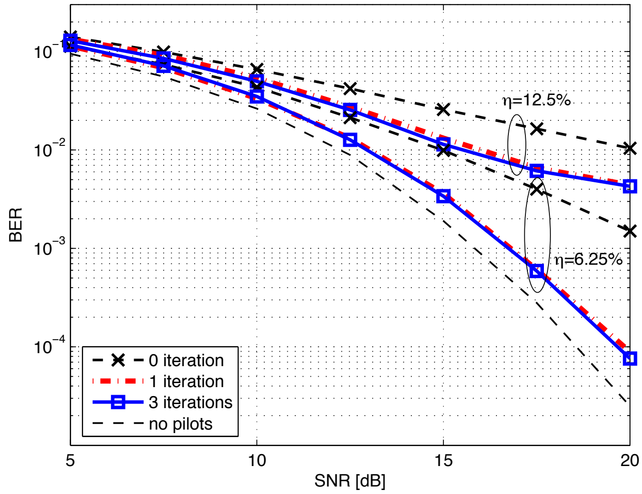

Fig. 4. BER Performance of FDSPT for Uncoded System using Linear MMSE Equalizer with Iterative Decision-directed Interference Cancellation, Known CSI

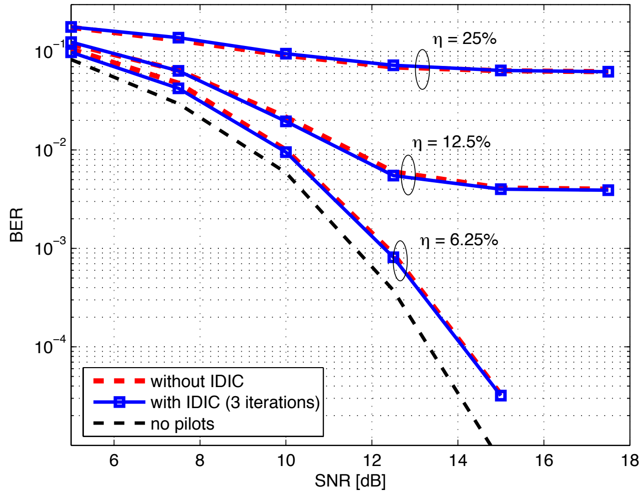

Fig. 5. BER Performance of FDSPT ($\alpha = 0$) for Uncoded System using IBSDFE Equalizer with Known CSI (IDIC: iterative decision-directed interference cancellation)

in IBSDFE, Y_ℓ is forced to zero for pilot tones, since there is no data there (or $\alpha = 0$).

Fig. 4 shows the BER performance of an uncoded QPSK system using FDSPT pilots ($\alpha = 0$) in the WINNER C2 channel with iterative decision-directed interference cancellation. The system parameters are given in Table I (see Sec. V), except for the last four rows that apply only to simulations with channel coding. The CSI is assumed to be known, i.e. $\sigma^2_{\Delta H_\ell} = 0$. Note that feedback data decisions are not necessarily correct; i.e. $\sigma^2_{\Delta A_\ell} \neq 0$. [8] Different values of η are obtained by fixing M and varying N. Pilots are added in every block. A linear MMSE equalizer is used to show the performance of the iterative interference cancellation technique. For $\eta = 6.25\%$, the BER performance is about 0.8 dB away from the one without FDSPT pilots. The irreducible BER degradation, resulting from the data decision errors, is more obvious for larger values of η, such as $\eta = 12.5\%$. Moreover, one iteration gives similar performance to that of three iterations.

When the IBSDFE equalizer is used, the performance improvement using iterative decision-directed interference cancellation is less significant, as shown in Fig. 5. Note that the same system parameters as those used in Fig. 4 are used, except that the IBSDFE equalizer is used instead of the linear MMSE equalizer. From Fig. 5, we can also see that when $\eta = 25\%$, i.e. one out of four data tones is removed, an error floor occurs at BER$= 6 \times 10^{-2}$. When a lower pilot overhead ratio, such as $\eta = 6.25\%$, is used the SNR degradation is about 0.8 dB at BER$=1 \times 10^{-4}$. Given a relatively large value of η, e.g. 25%, for a particular pilot block, an alternative approach to reduce the error floor is to use channel coding over a frame with sparse pilot blocks. This is shown in the simulation results in Sec. V. Note that related results are shown in [18], [19], but with lower pilot overhead ratios.

[8]Known CSI is used in Fig. 4 and Fig. 5 to show the effects of the interference due to FDSPT pilots, decoupling the effects of channel estimation errors.

D. Complexity and Spectral Efficiency

Assuming that the number of data symbols M is the same for SC systems with N FET and N FDSPT pilots, the implementation complexity for FET pilots and FDSPT pilots ($\alpha = 0$) is the same in the transmitter. In the receiver, additional N complex multiplications are needed for interference cancellation for FDSPT pilots ($\alpha = 0$). However, for FDSPT pilots ($\alpha \neq 0$), additional N complex multiplications and N complex additions are needed in the transmitter for superimposing pilot tones onto scaled data tones. In the receiver, additional $2N$ complex multiplications and $2N$ complex additions are needed for pilot removal and interference cancellation, assuming that no further processing is performed to remove interfering data tones at the pilot locations except for removing the scaled pilot tones. In general, FDSPT pilots with iterative decision-directed interference cancellation requires slightly higher complexity than FET pilots. However, SC systems with FDSPT pilots have better spectral efficiency as the adding of FDSPT pilots consume no extra bandwidth. In the case considered in this paper (where 256 pilots/block are present in 2 blocks within a frame of 12 blocks), FET has 4.3% pilot overhead, defined as the total number of pilots divided by total number of data symbols, while FDSPT has 0% pilot overhead. Note that higher pilot overhead is expected if pilots are present in more blocks within a frame.

IV. Iterative Frequency Domain Channel Estimation using In-Band Pilots

To use the tentative decisions to improve the channel estimates, the IDDCE described in [25] is modified for SC systems with FDM pilots. Instead of using the LLR of the coded symbols from the output of an APP decoder, the hard decisions either from the equalizer output or from the Viterbi decoder output are used for low complexity. This is suboptimum, but as we will see from the simulation results, it gives performance close to the case of known channel frequency response. Fig. 6 shows the structure of the proposed

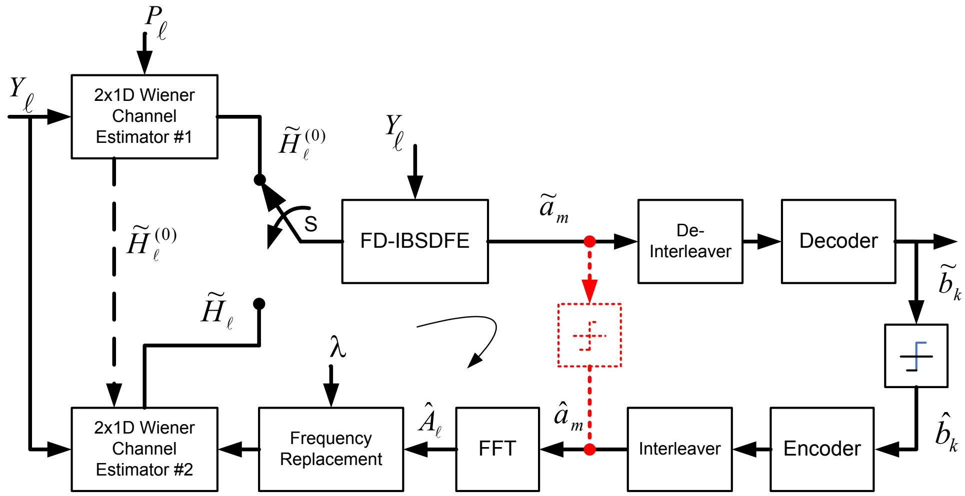

Fig. 6. Block Diagram of the Proposed Frequency Domain Iterative Decision-Directed Channel Estimator for SC Systems with or without Decoding Part

IDDCE for SC systems with or without involving the decoding process. The dotted hard decision block is used when the decoder is not involved, ignoring the interleaver/de-interleaver and encoder/decoder. During the initial channel estimation using cascaded 2×1D Wiener filter #1, the switch S is in the upper position. Note that in obtaining the cascaded 2×1D Wiener filter taps we assume a robust channel estimator which has a uniform (rectangular) power delay profile and a uniform Doppler spectrum [30], [31], of which the statistics may be mismatched to a wide range of power delay profiles and Doppler spectrum. Using a robust channel estimator has the advantage of requiring no *a priori* knowledge of the actual power delay and Doppler power spectrum of the channel, by assuming worse case scenarios of the power delay profile and Doppler power spectrum. Moreover, we assume the actual maximum echo delay, the vehicle speed and the noise variance are known at the receiver. In the frequency direction, the *nearest* sixteen pilots, in terms of absolute distance, are used for interpolation, while in the time direction we assume two pilots for interpolation.

The initial estimated channel frequency response $\tilde{H}_\ell^{(0)}$ is then used by the IBSDFE, producing the soft decision of the transmitted data symbol $\tilde{a}_m$. If the decoder is not involved, the hard decision of $\tilde{a}_m$, denoted as $\hat{a}_m$, is then used to calculate $\hat{A}_\ell$ by taking the FFT of $\hat{a}_m$. If the decoder is involved, after de-interleaving and soft Viterbi decoding, the hard decisions of the decoder output $\hat{b}_k$ are *re-encoded* and interleaved to obtain the hard decisions of the transmitted data symbols $\hat{a}_m$. Given the received frequency sample Y_ℓ and $\hat{A}_\ell$, our next task is to find the least square (LS) estimates of the channel frequency responses for the Wiener filtering process, given as [32], [33]

$$\hat{H}_\ell = \frac{Y_\ell}{\hat{A}_\ell}, \quad \ell = 0, 1, \ldots, M-1 \tag{21}$$

where $\hat{A}_\ell$ is the FFT of $\hat{a}_m$, given as

$$\hat{A}_\ell = \frac{1}{\sqrt{M}} \sum_{m=0}^{M-1} \hat{a}_m e^{-\frac{j2\pi m\ell}{M}}, \quad \ell = 0, 1, \ldots, M-1. \tag{22}$$

A. MMSE Estimator

Using the Central Limit Theorem [34], $\hat{A}_\ell$ in (22) can be approximated as a Gaussian random variable with zero mean and a variance of σ_a^2. From (21), assuming all the decisions from the equalizer are correct, the variance of $\hat{H}_\ell$ given $\hat{A}_\ell$ can be found as

$$\hat{\sigma}_\ell^2 = E\{|H_\ell - \hat{H}_\ell|^2\} = \frac{\sigma^2}{|\hat{A}_\ell|^2} \tag{23}$$

where σ^2 is the noise variance. It is obvious that from (23), when $|\hat{A}_\ell|$ is small, *noise enhancement* makes interpolation of $\hat{H}_\ell$ to find $\tilde{H}_\ell$ difficult.

Using (21) to find the LS (or *raw*) estimates for the second Wiener filter can result in the issue of noise enhancement. Our first approach to overcome this issue is to use an optimum MMSE estimator to find the raw estimates. Assuming that all the decisions are correct so that $\hat{A}_\ell = A_\ell$ for $\ell = 0, 1, \cdots, M-1$, the received frequency samples can be written, in matrix form, as

$$\mathbf{Y} = \hat{\mathbf{A}}\mathbf{H} + \mathbf{V} \tag{24}$$

where $\hat{\mathbf{A}}$ is a diagonal matrix with $[\hat{A}_0, \hat{A}_1, \cdots, \hat{A}_{M-1}]^T$ being the main diagonal elements, $\mathbf{H} = [H_0, H_1, \cdots, H_{M-1}]^T$ is the channel frequency response and $\mathbf{V} = [V_0, V_1, \cdots, V_{M-1}]^T$ is the noise frequency samples. Since $\mathbf{Y}$ and $\mathbf{H}$ are jointly complex Gaussian, the MMSE (conditional mean) estimator for $\mathbf{H}$, given $\mathbf{Y}$ and $\hat{\mathbf{A}}$, is given as [33, pp. 533],

$$\hat{\mathbf{H}}' = \mathbf{C}_{HH}\hat{\mathbf{A}}^H \left(\hat{\mathbf{A}}\mathbf{C}_{HH}\hat{\mathbf{A}}^H + \mathbf{C}_{VV}\right)^{-1}\mathbf{Y} \tag{25}$$

where ^{H} denotes the Hermitian transpose, $\mathbf{C}_{HH}$ and $\mathbf{C}_{VV} = \sigma^2\mathbf{I}$ are the covariance matrix of the channel frequency response and the noise samples, respectively. Inspecting (25), we notice that to obtain $\hat{\mathbf{H}}'$ not only requires the knowledge of the covariance matrix of the channel frequency response $\mathbf{C}_{HH}$ but also the DFT of the data decision $\hat{\mathbf{A}}$, which varies from block to block. This means that we need to calculate the filter taps on-line for each block. Note that by dividing $\mathbf{Y}$ into smaller size vectors, a smaller size of matrix inversion can be used to reduce the complexity, at the expense of

TABLE I
SIMULATION SYSTEM PARAMETERS [35]

Parameter	Values
Carrier frequency [GHz]	3.7
Duplex Mode	FDD
System bandwidth [MHz]	40.0
Number of subcarriers in use (M)	1024
Subcarrier spacing [Hz]	39062.5
Cyclic prefix length [μs]	3.2
Total symbol length [μs]	28.8
TD windowing rolloff factor[%]	11
Modulation	QPSK
Upsampling factor	5
Pilot spacing in subcarries (K)	4
Maximum Doppler Frequency [Hz]	171.3
Number of blocks per frame	12
Blocks within a frame with pilots	1st,12th
Channel coding (convolutional)	$(133, 171)_O$

slight performance degradation, as shown in Sec. V-B. (25) provides an optimum solution with relatively high complexity for practical implementation of the proposed iterative channel estimator.

B. Sub-optimum Low Complexity Technique: The Frequency Replacement Algorithm

To overcome the issue of high complexity in (25) and the issue of noise enhancement, we propose to replace the frequency channel estimates at *noise enhanced* frequencies with the initial channel frequency estimates (or the channel estimates in the previous iteration) using a threshold λ. We called this method *frequency replacement* (FR). The *mixed* LS estimates using tentative decision tones and the channel frequency estimates in the previous iteration are given as

$$\hat{H}'_\ell = \begin{cases} \frac{Y_\ell}{\hat{A}_\ell}, & |\hat{A}_\ell| > \lambda \\ \tilde{H}^{prev}_\ell, & |\hat{A}_\ell| \leq \lambda \end{cases} \tag{26}$$

where $\ell = 0, 1, \ldots, M-1$ and $\tilde{H}^{prev}_\ell$ is the channel frequency estimates in the previous iteration. Since $\hat{A}_\ell$ can be modeled as a Gaussian random variable, $|\hat{A}_\ell|$ has a Rayleigh probability density function. Hence, the probability of replacing the frequencies can be calculated as

$$Pr\{|\hat{A}_\ell| \leq \lambda\} = 1 - e^{-\lambda^2} \tag{27}$$

where we assume $E\{|\hat{A}_\ell|^2\} = 1$. Assuming that $Pr\{|\hat{A}_0| \leq \lambda\} = Pr\{|\hat{A}_1| \leq \lambda\} = \ldots = Pr\{|\hat{A}_{M-1}| \leq \lambda\}$, the average number of frequencies replaced for a particular block can be calculated as

$$N_{FR} = M\left(1 - e^{-\lambda^2}\right) \tag{28}$$

where M is the number of frequencies per block. It should be noted that the value of the threshold λ determines the number of frequencies to be replaced. When λ is too low, noise enhancement persists. In the other hand, if λ is too high, most of the channel frequency estimates in the previous iteration will be re-used, resulting in similar performance as that of the previous channel estimates. Assuming that $E\{|\hat{A}_\ell|^2\} = 1$, the threshold value should be chosen between 0 and 1, i.e. $0 < \lambda < 1$. In this paper, the threshold value λ is found by simulation. The FR algorithm is a low complexity technique, as no matrix inversion is required.

TABLE II
URBAN MACRO POWER DELAY PROFILE [2]

Power[dB]	-0.5 0 -3.4 -2.8 -4.6 -0.9 -6.7 -4.5 -9.0 -7.8 -7.4 -8.4 -11.0 -9.0 -5.1 -6.7 -12.1 -13.2 -13.7 -19.8
Delays[ns]	0 5 135 160 215 260 385 400 530 540 650 670 720 750 800 945 1035 1185 1390 1470

C. Wiener $2 \times 1D$ Filtering of Raw Estimates

Using the frequency channel estimates from either (25) or (26), another $2 \times 1D$ Wiener filter (similar to the one used for initial channel estimation) with the smallest *pilot* distance in both time and frequency direction (denoted as D_t and D_f, respectively), i.e. $D_f = D_t = 1$, is used to refine the channel estimates for the whole frame. Note that denser pilots for Wiener filtering in general gives better smoothing performance. In this case, we use the sixteen nearest *pseudo pilots* for smoothing in the frequency direction, while we use Q (number of blocks per frame) *pseudo pilots* for smoothing in the time direction. Switch S in Fig. 6 is then in the lower position, allowing the improved channel frequency estimates to be used by the IBSDFE and forming a closed loop for the next iteration. The difference between the $2 \times 1D$ Wiener filter #1 and $2 \times 1D$ Wiener filter #2 in Fig. 6 is that the values of D_f and D_t in $2 \times$ 1D Wiener filter #2 are always 1, while the corresponding values for $2 \times 1D$ Wiener filter #1 are $D_t = Q > 1$ and $D_f = K > 1$. In our simulation, we assume $Q = 12$ and $K = 4$ and the pilots are added in the first and the last block of the frame.

V. SIMULATION RESULTS

Table I shows the system parameters [35] for our simulation. The number of samples per block, excluding CP, is $L = 5120$. Channel coding is used over a frame with 12 blocks. Note that the FET pilot overhead is $2 \times 256/(1024 \times 12 - 2 \times 256) = 4.3\%$. A random bit interleaver over a frame is also included. Table II shows the power delay profile of the WINNER C2 (urban macro-cell) channel [2]. Jakes' Doppler spectrum was assumed to generate channel realizations with user terminal speed of 50 km/h. The channel is assumed to be static during one OFDM symbol and changes according to Jakes model from one OFDM symbol to another OFDM symbol. The soft Viterbi algorithm is used in the decoder.

A. Determination of the Best Value of the Threshold λ

We first determine the *best* value of λ by simulating a SC system with IDDCE using different values of λ at wide range of SNR. Fig. 7 shows the average MSE of the IDDCE using the FR algorithm with various values of λ at SNR = 5, 10, and 20 dB. For all three SNR values, the best (in terms of smallest MSE by observation) values for FR algorithm was found to be around $\lambda = 0.3$. With $\lambda = 0.3$ and using (27), the probability of replacing a particular frequency using the FR algorithm is

$$Pr\{|\hat{A}_\ell| \leq \lambda\} = 1 - e^{-(0.3)^2} = 8.6\% \tag{29}$$

and the average number of frequencies replaced in a block is, using (28), $N_{FR} = M \times 8.6\% \approx 89$. This means that we are using most (91.4%) of the tentative decision tones for channel estimation for the next iteration.

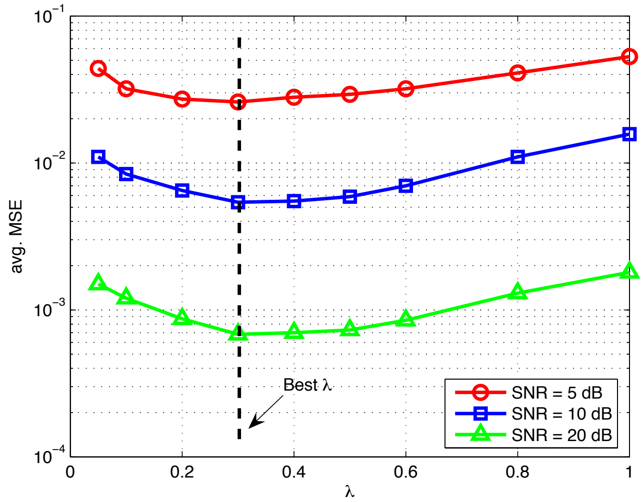

Fig. 7. Effects of the Threshold Values for IDDCE

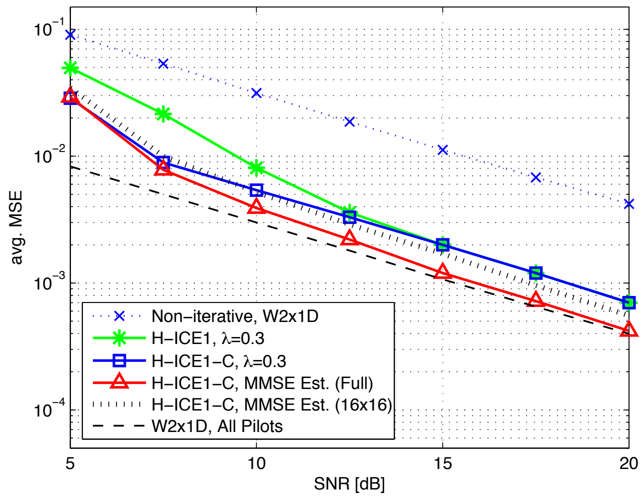

Fig. 8. MSE Performance of IDDCE using FET Pilots

B. MSE Performance

Fig. 8 depicts the averaged (over all frequencies in a frame) MSE for the IDDCE with FET pilots. The MSE performance curve for non-iterative W2x1D and a frame with all Chu pilots are also included for comparison. The MSE performance of using all Chu pilots can be considered as a lower bound on MSE for IDDCE since there exist no noise enhanced raw estimates and decision errors. The legend 'H-ICE1, $\lambda = 0.3$' means hard decision iterative channel estimation with 1 iteration and FR algorithm with $\lambda = 0.3$ is used. The 'C' in the legend 'H-ICE1-C, $\lambda = 0.3$' means the decoder output decisions are included in the channel estimation process. We see from Fig. 8 that IDDCE with the FR algorithm in general gives MSE performance improvement over the non-iterative cascaded 2x1D Wiener filter. However, in the low SNR region from 5 dB to 12 dB, the MSE performance of the IDDCE with FR algorithm without involving the decoder is degraded as a result of decision errors. The IDDCE using the FR algorithm with re-encoding improves the MSE performance in the low SNR region. The use of the optimum MMSE

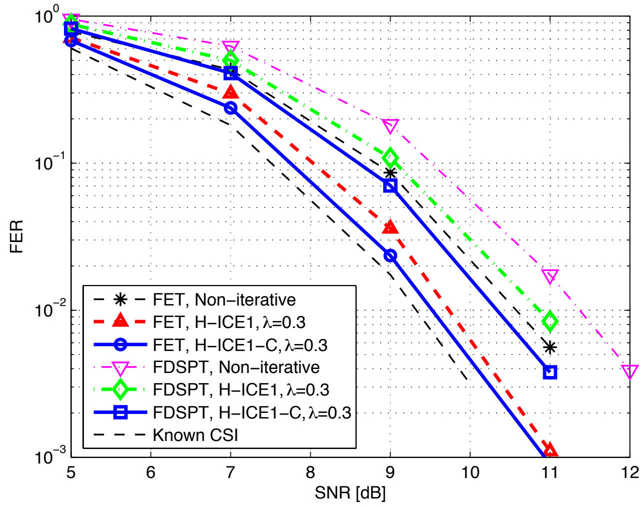

Fig. 9. FER Performance of Coded SC Systems with FDM Pilots using the Proposed IDDCE and IBSDFE Equalizer

estimator to obtain the raw estimates for the second Wiener filtering gives MSE performance close to that of using all Chu pilots in the high SNR region ($>$ 12 dB), where decision errors are small. On the other hand, in the low SNR region ($\leq$ 10 dB), the low complexity IDDCE using the FR algorithm with a threshold gives comparable MSE performance to that of using the optimum MMSE estimator. Moreover, when a smaller size of matrix inversion, e.g. 16×16, is used, the MSE performance is similar to that of using the proposed low complexity IDDCE.

C. Frame Error Rate Performance

Fig. 9 shows the frame error rate (FER) performance of the coded SC systems with FDSPT ($\alpha = 0$) and FET pilots using the proposed IDDCE technique using the FR algorithm wiht $\lambda = 0.3$. Note that the IBSDFE equalizer is used. Comparing with the non-iterative 2×1D Wiener interpolation for FET pilots, the IDDCE without decoding in the loop improves about 1 dB for FER=10^{-2} and about 1.2 dB when decoding is included in the iteration. This is about 0.2 dB away from the known CSI case. The use of IDDCE for FDSPT ($\alpha = 0$) pilots improves about 1 dB for FER=10^{-2} with decoding in the loop and about 0.5 dB with decoding not in the loop, compared with that of non-iterative 2×1D Wiener filtering. For FDSPT ($\alpha = 0$), the FER performance is about 1 dB, at FER=10^{-2}, away from the known CSI case.

For OFDM systems with FET pilots, shown in Fig. 10, the improvement for IDDCE without decoding over the non-iterative estimator is about 0.2 dB, while the improvement for IDDCE with decoding is about 1 dB, which is about 0.2 dB away from the known CSI case. Similar to the SC case, using FDSPT pilots for the OFDM system requires about 1 dB extra SNR at a FER of 10^{-2}. Finally, Fig. 11 shows the FER for coded SC system using FDSPT ($\alpha = 0$) pilots with iterative decision-directed interference cancellation, where 'IDIC1' means iterative decision-directed interference cancellation with 1 iteration. When decoding is included, the improvement is about 0.5 dB, which is about 0.3 dB from

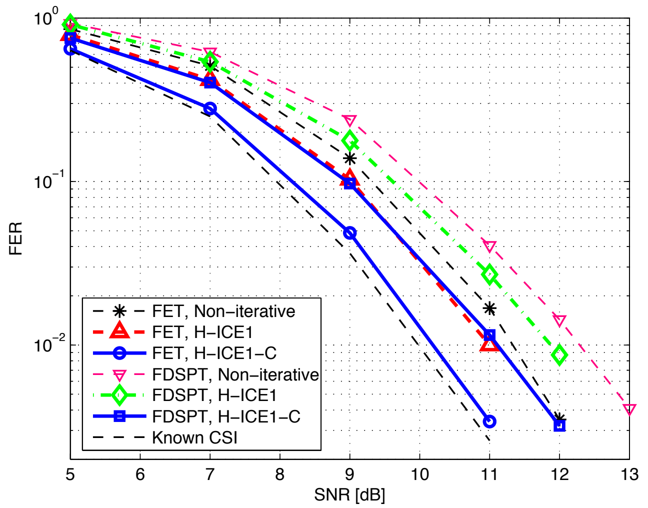

Fig. 10. FER Performance of Coded OFDM Systems with FDM Pilots using the IDDCE Proposed in [25] and Linear Equalizer

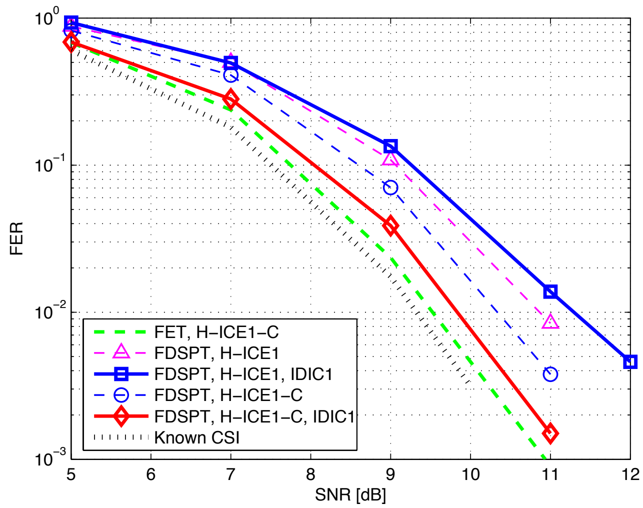

Fig. 11. FER Performance of FDSPT for Coded SC Systems using IBSDFE Equalizer with Iterative Decision-directed Interference Cancellation, with Channel Estimation

FET pilots and 0.5 dB away from the performance for known CSI. In other words, using the proposed iterative decision-directed interference cancellation technique, the improvement is about 0.5 dB at FER = 10^{-2}. Notice that when decoding is not used in the loop, a performance degradation of about 0.4 dB is observed, resulting from the less reliable data decisions for interference cancellation.

VI. Conclusions

Two techniques of FDM pilot insertion have been presented: FDSPT, where pilots are superimposed on scaled or replace data-carrying tones and FET, where groups of data carrying tones are shifted for multiplexing of pilot tones. Using FDM pilots in a SC system facilitates flexible and efficient assignment of signals to available spectrum. A slightly modified form of the IBSDFE algorithm was introduced. It and the IBSDFE algorithm allowed the FDSPT pilot system to approach the performance of FET pilots to within less than 0.5 dB.

We also proposed a low complexity IDDCE technique, modified from the channel estimator described in [25], for SC systems with FDM pilots. The tentative decisions are used as extra *pilots* to further improve the initial frequency channel estimates using a cascaded 2×1D Wiener filter. The issue of noise enhancement when finding the raw estimate of the channel frequency response using the tentative decisions can be overcome by an MMSE estimator, which has relatively high complexity. To further reduce the complexity, we proposed to use the FR algorithm, which replaces the noise enhanced *raw* estimates of the channel frequency response with the corresponding channel frequency estimates in the previous iteration, by comparing with a threshold. Using the proposed low complexity IDDCE with FR algorithm, the FER performance for coded SC systems using FET pilots and FDSPT pilots with iterative interference cancellation was found to be about 0.2 dB and about 0.5 dB away from the FER performance with known channel frequency response at FER=10^{-2}, respectively. FDSPT pilots can also be used for OFDM systems with channel coding. It was found that an extra 1 dB of SNR is required at FER 10^{-2}, comparing with that using the conventional FET pilots for OFDM systems. Note however that FDSPT pilots requires no pilot overhead.

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References

[1] J. Korhonen, *Introduction to 3G Mobile Communications*, Norwood, U.S.A., Artech House Inc., 2nd Edition, 2003.

[2] E. Zimmermann, et al., "Final report on identified RI key technologies, system concept, and their assessment," in *WINNER Deliverable D2.10*, Dec. 2005. Available: https://www.ist-winner.org/DeliverableDocuments/D2.10.pdf

[3] TS36.212, "Physical channels and modulation," in *3GPP TSG RAN*, v. 1.0.0, Mar. 2007.

[4] J. Coon, M. Sandell, M. Beach and J. McGeehan, "Channel and noise variance estimation and tracking alogithms for unique-word based single-carrier systems," *IEEE Trans. Wireless Comm.*, vol. 5, no. 6, pp. 1488-1496, Jun. 2006.

[5] X. Ma, G. Giannakis, and S. Ohno, "Optimal training for block transmissions over doubly selective wireless fading channels," *IEEE Trans. Signal Processing*, vol. 51, no. 5, pp. 1351-1366, Nov. 2003.

[6] D. Falconer, S. L. Ariyavisitakul, A. Benyamin-Seeyar, and B. Eidson, "Frequency domain equalization for single-carrier broadband wireless systems," *IEEE Trans. Commun.*, vol.40, pp. 58-66, Apr. 2002.

[7] L. Deneire, B. Gyselinckx, and M. Engels, "Training sequence versus cyclic prefix-a new look on single carrier communications," *IEEE Commun. Letters*, vol.5, pp. 292-294, Sep. 2001.

[8] J. Cavers, "An analysis of pilot symbol assisted modulation for Rayleigh fading channels," *IEEE Trans. Veh. Technol.*, vol. 40, pp. 686-693, Nov. 1991.

[9] P. Hoher, "TCM on frequency selective land-mobile radio channels," in *Proc. 5th Tirrenia Int. Workshop on Dig. Commun.*, pp. 317-328, Sep. 1991.

[10] P. Hoher, S. Kaiser, and P. Robertson, "Pilot-symbol-aided channel estimation in time and frequency," in *Proc. Communication Theory Mini-Conference (CTMC) within IEEE GLOBECOM'97*, pp. 90-96, 1997.

[11] Z. Wang and G. Giannakis, "Wireless multicarrier communications where Fourier meets Shannon, *IEEE Signal Processing Mag.*, vol. 17, pp. 2948, May 2000.

[12] R. Tafazolli (Editor), *Technologies for the Wireless Future - Wireless World Research Forum (WWRF)* , West Sussex, England, John Wiley & Sons Ltd., vol. 2, 2006.

[13] R. Dinis, D. Falconer, C. T. Lam and M. Sabbaghian, "A multiple access scheme for the uplink of broadband wireless systems," in *Proc. IEEE GLOBECOM Conf.*, vol. 6, pp. 3808-3812, Dec. 2004.

[14] U. Sorger, et al., "Interleaved FDMA - a new spread-spectrum multiple-access scheme," in *Proc. International Communications Conference within IEEE ICC'98*, vol.2, pp. 1013-1017, 1998.

[15] D. Chu,"Polyphase codes with good periodic correlation properties," *IEEE Trans. Inform. Theory*, pp. 531-532, Jul. 1972.

[16] J. P. McGeehan and A. J. Bateman,"Phase-locked transparent tone-in-band (TTIB): A new spectrum configuration particularly suited to the ransmission of data over ssb mobile radio networks," *IEEE Trans. Commun.*, vol.32, pp. 81-87, Jan. 1984.

[17] F. Davarian,"Mobile digital communications via tone calibration," *IEEE Trans. Vehicular Technol.*, vol. 36, pp. 55-62, May 1987.

[18] M. Ghogho, D. McLernon, E. A.-Hernandez and A. Swami,"Channel estimation and symbol detection for block transmission using data-dependent superimposed training," *IEEE Signal Processing Lett.*, vol. 12, no.3, pp. 226-229, Mar. 2005.

[19] J. K. Tugnait and S. He, "Doubly-selective channel estimation using data-dependent superimposed training and exponential bases models," in *Proc. 40th Annual Conf. on Information Sciences and Systems*, 2006.

[20] S. Ohno and G. Giannakis, "Optimal training and redundant precoding for block transmissions with application to wireless OFDM," *IEEE Trans. Commun.*, vol. 50, no. 12, pp. 2113-2123, Dec. 2002.

[21] J. K. Tugnait and X. Meng, "Superimposed training for channel estimation: performance analysis, training power allocation, and frame synchronization," *IEEE Trans. Signal Processing.*, vol. 54, no. 2, pp. 752-765, Feb. 2006.

[22] F. Chin, B. Farhang-Boroujeny and C. Ho, "Added pilot semi-blind channel estimation scheme for OFDM in fading channels," in *Proc. IEEE Globecom*, vol. 5, pp. 3075-3079, 2001.

[23] H. Zhu, B. Farhang-Boroujeny and C. Schlegel, "Pilot embedding for joint channel estimation and data detection in MIMO communication systems," *IEEE Commun. Lett.*, vol. 7, no. 1, pp. 30-32, Jan. 2003.

[24] H.-J. Su, and E. Geraniotis, "Low-complexity joint channel estimation and decoding for pilot symbol-assisted modulation and multiple differential detection system with correlated Rayleigh fading," *IEEE Trans. Commun.*, vol. 50, no. 2, pp. 249-261, Feb. 2002.

[25] F. Sanzi, S. Jelting, and J. Speidel, "A comparative study of iterative channel estimators for mobile OFDM systems," *IEEE Trans. Wireless Commun.*, vol. 2, no. 5, pp. 849-858, Sep. 2003.

[26] B. Ng, C.-T. Lam, and D. Falconer, "Turbo frequency domain equalization for single-carrier broadband wireless systems," *IEEE Trans. Wireless Commun.*, vol. 6, no. 2, Feb. 2007.

[27] L. Davis, I. Collings, and P. Hoeher, "Joint MAP equalization and channel estimation for frequency-selective and frequency-flat fast-fading channels," *IEEE Trans. Commun.*, vol. 49, no. 12, pp. 2106-2114, December 2001.

[28] N. Benvenuto and S. Tomasin, "Iterative design and detection of a DFE in the frequency domain," *IEEE Trans. Commun.*, vol. 53, no. 11, pp. 1867-1875, Nov. 2005.

[29] C.-T. Lam, D. Falconer, F. Danilo-Lemoine and R. Dinis, "Channel estimation for SC-FDE systems using frequency domain multiplexed pilots," in *Proc. IEEE VTC 2006*, pp. 1-5, Sep. 2006.

[30] G. Auer, "Channel estimation for OFDM systems with multiple transmit antennas by filtering in time and frequency," in *Proc. IEEE GLOBECOM '03*, vol. 1, pp. 322-326, Dec. 2003.

[31] Y. Li, L. Jr. Cimini, and N. Sollenberger,"Robust channel estimation for OFDM systems with rapid dispersive fading channels," *IEEE Trans. Commun.*, vol. 46, no. 7, pp. 902-915, Jul. 1998.

[32] L. Hanzo, M. Münster, B. J. Choi and T. Keller, *OFDM and MC-CDMA for broadband ulti-user communications, WLANs and broadcasting*, West Sussex, England, John Wiley & Sons, Ltd., 2003.

[33] S. Kay, *Fundamentals of statistical signal processing : estimation theory*, Englewood Cliffs, N.J., Prentice-Hall, 1993.

[34] A. Papoulis, *Probability, Random Variables, and Stochastic Processes*, 3rd Edition, Singapore, McGraw-Hill Inc., 1991.

[35] G. Vivier, et al., "WINNER II test scenarios and calibration cases issue 1," in *WINNER Deliverable D6.13.1*, Jun. 2006. Available: https://www.ist-winner.org/WINNER2-Deliverables/D6.13.1.pdf

Chan-Tong Lam received the B.Sc. (Eng) and M.Sc. (Eng) degrees in 1998 and 2000, respectively, both from Queen's University, Kingston, Canada and the Ph.D. degree in 2007 from Carleton University, Ottawa, Canada. He worked for Sigpro Wireless Inc. in Ottawa, Canada, from November 2000 to May 2003, as a 3rd Generation (3G) Systems Engineer. While pursuing his PhD in the Department of Systems and Computer Engineering at Carleton University, he also participated in the European Wireless World Initiative New Radio (WINNER) project (within the work package of physical layer design) from January 2004 to August 2007 as a research assistant. He is now with the Bureau of Telecommunications of Macao as a Technical Officer and he is also a part-time lecturer at the Macao Polytechnic Institute. His research interests include wireless communications and networks and digital signal processing for biomedical engineering.

David Falconer received the B.A. Sc. degree in Engineering Physics from the University of Toronto in 1962 and the S.M. and Ph.D. degrees in Electrical Engineering from M.I.T. in 1963 and 1967 respectively. After a year as a postdoctoral fellow at the Royal Institute of Technology, Stockholm, Sweden he was with Bell Laboratories, Holmdel, New Jersey from 1967 to 1980, as a member of the technical staff and later as group supervisor. During 1976-77 he was a visiting professor at Linköping University, Linköping, Sweden. Since 1980 he has been with Carleton University, Ottawa, Canada, where he is now Professor Emeritus and Distinguished Research Professor in the Department of Systems and Computer Engineering. His current research interests center around beyond-third-generation broadband wireless communications systems. He was awarded Communications Society Prize Paper Awards in Communications Circuits and Techniques in 1983 and 1986. He was a co-recipient of the IEEE Vehicular Technology Transactions best paper of the year award in 1992. From 1981 to 1987 he was Editor for Digital Communications of the IEEE Transactions on Communications. He was Director of Carleton's Broadband Communications and Wireless Systems (BCWS) Centre from 2000 to 2004. He was the Chair of Working Group 4 (New Radio Interfaces, Relay-Based Systems and Smart Antennas) of the Wireless World Research Forum (WWRF) in 2004 and 2005. He is an IEEE Life Fellow.

Florence Danilo-Lemoine received the engineering degree in Electrical Engineering from the National School of Electronics and Radio-Electricity (E.N.S.E.R.G.), Grenoble, France, in 1992, and the Ph.D. degree in Electrical Engineering from McGill University, Montreal, Canada, in 2000. From 1992 to 2000, she was a teaching and research assistant, and held several part-time lecturer positions at McGill University. In 1999, she was a faculty lecturer in the Department of Electrical Engineering at McGill University. In 2000, she joined the Faculty of Engineering and Design at Carleton University as an assistant professor in the Department of Systems and Computer Engineering, Ottawa, Canada. Since July 2007, she has been an adjunct professor in the Department of Systems and Computer Engineering at Carleton University. Her research interests are in digital telecommunications, detection and statistical signal processing with emphasis on wireless communication systems. In 1999, she was awarded the Canadian Advanced Technology alliance (CATA) telecommunications graduate scholarship, and in 2001 she received a university faculty award from the Natural Sciences and Engineering Research Council of Canada (NSERC).